Math Inquirers

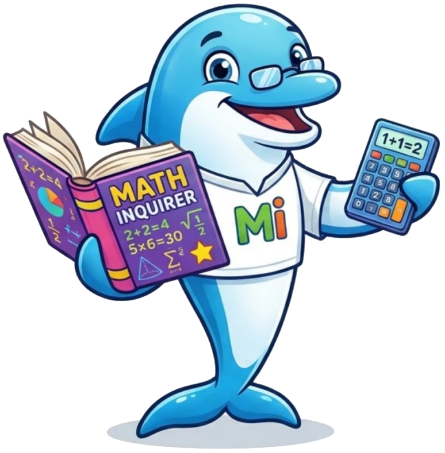
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Tutor Answer Key

Book 1

Lessons 1 to 5



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

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Math Inquirers Level 1, Term 1 - Tutor Answer Key, Book 1

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Answer key: Lesson 1, Bundles and Singles

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Count the sticks.

Answer: 2 bundles, 6 singles, 26 sticks

Count the 2 bundles in tens: 10, 20. Then count on the 6 singles: 21, 22, 23, 24, 25, 26. So 2 bundles, 6 singles, 26 sticks.

Watch for: Counting each bundle as 1 (2 + 6 = 8).; Writing the digits the wrong way round (62).

S2. Draw 5 single sticks and 3 bundles of ten in the box.

Answer: 35 (3 long lines for the bundles and 5 short lines for the singles)

3 long lines are 3 bundles of ten (that is 30), and 5 short lines are 5 singles. Count: 10, 20, 30, 31, 32, 33, 34, 35.

Watch for: Drawing 3 sticks and 5 sticks instead of 3 bundles of ten.; Writing 53.

S3. Fill in the missing numbers.

Answer: 42, 70, 19, 85

(a) 4 tens is 40, and 2 more is 42. (b) 7 tens is 70 and there are no ones: 70. (c) 10 and 9 is 19. (d) 8 tens is 80, and 5 ones makes 85.

Watch for: Writing 7 for 7 tens and 0 ones.; (d) writing 58 because 5 was said first.

S4. Fill in the blanks.

Answer: 5 tens 8 ones; 4 tens 0 ones; 7 tens 1 one

(a) 58: the 5 is the tens digit, the 8 is the ones digit. (b) 40: 4 tens and no ones. (c) 71: 7 tens and 1 one.

Watch for: Saying 40 has 4 ones.; Swapping the two digits.

S5. Ali has 47 sticks. He ties the sticks into bundles of 10. Fill in the blanks.

Answer: 4 bundles, 7 left over

47 is 4 tens and 7 ones, so Ali makes 4 full bundles and has 7 sticks left over (4 bundles hold 40 sticks).

Watch for: Answering 47 for the bundles.; Saying 3 bundles and 17 left over: 17 singles can still make another bundle.

S6. Kumar has 3 bundles of ten and 4 single sticks. He writes 43.

Answer: No. 34. He wrote the tens and ones digits the wrong way round (43 is 4 tens and 3 ones).

3 bundles is 30 and 4 singles makes 34. Kumar wrote 43, which is 4 tens and 3 ones. The tens digit goes on the left.

Watch for: Saying he is right because the digits 3 and 4 are there.; Not saying what he mixed up (any clear sentence about the digits is fine).

S7. Ali has 6 bundles of ten and 0 single sticks. Mei has 0 bundles and 6 single sticks. Draw Ali's sticks, then Mei's sticks, in the box. Then write each number.

Answer: Ali 60, Mei 6. Ali has more (6 tens is 60; 6 singles is 6)

Ali: 6 bundles of ten is 60, with no singles, so 60. Mei: 6 single sticks is 6. The same digit 6 means very different amounts because of its place: 6 tens or 6 ones.

Watch for: Writing 6 for both children.; Drawing 6 short lines for Ali instead of 6 long lines (6 bundles of ten).

Think harder (page 3)

T1. Mei has 5 bundles of ten and 14 loose sticks. How many sticks does she have?

Answer: 64 (14 loose sticks make 1 more bundle and 4 left, so 6 tens and 4 ones)

14 loose sticks = 1 bundle and 4 singles. So Mei has 5 + 1 = 6 bundles and 4 singles: 64 sticks. Check: 50 + 14 = 64.

Watch for: Writing 514 by joining the digits.; Writing 5 tens and 14 ones as 5 + 14 = 19.

T2. Fill in the blank.

Answer: 11 (3 tens is 30, and 30 + 11 = 41)

Opening a bundle gives 10 more singles. 1 single + 10 singles = 11 singles, and 3 bundles are left. 30 + 11 = 41, so it is the same number.

Watch for: Writing 10 (forgetting the 1 single already there).; Writing 1 (forgetting the opened bundle).

T3. My tens digit is more than 3. My ones digit is 6. I am between 30 and 50. What number am I?

Answer: 46

Between 30 and 50 and ending in 6: 36 and 46. 36 has 3 tens, which is not more than 3. 46 has 4 tens, which is more than 3. So the number is 46.

Watch for: Stopping at 36 and not checking clue 3.; Listing numbers that start with 6.

T4. Mei has 7 tens and 2 ones. Ali has 6 tens and 12 ones. Circle one. Then show how you know.

Answer: The same: 72 each (6 tens 12 ones is 7 tens 2 ones)

Ali's 12 ones make 1 more bundle and 2 ones. So Ali has 7 tens and 2 ones, the same as Mei. Both have 72.

Watch for: Saying Ali has more because 12 is greater than 2.; Adding 6 + 12 = 18 and comparing 18 with 9.

Go further (page 4)

G1. Ben has 6 bundles of ten and 14 loose sticks. Siti has 5 bundles of ten and 29 loose sticks.

Answer: Siti. Ben has 74, Siti has 79, so Siti has 5 more.

Ben: 6 tens and 14 ones is 7 tens and 4 ones, so 74. Siti: 5 tens and 29 ones is 7 tens and 9 ones, so 79. 79 is 5 more than 74.

Watch for: Comparing only the bundles (6 and 5) and saying Ben has more.; Not opening the loose sticks into bundles before comparing.

G2. Pencils come in boxes of 10. Mei needs 60 pencils. She has 4 full boxes and 18 loose pencils.

Answer: 2 more pencils (4 boxes and 18 pencils is 58)

4 boxes is 40 pencils. 40 + 18 = 58. She needs 60, so she needs 60 - 58 = 2 more.

Watch for: Saying 18 more (the loose pencils).; Forgetting the 4 full boxes hold 40 pencils.

G3. Write 30 in different ways, like 3 tens and 0 ones.

Answer: 4 ways: 3 tens 0 ones; 2 tens 10 ones; 1 ten 20 ones; 0 tens 30 ones

3 tens 0 ones. Open one bundle: 2 tens 10 ones. Open another: 1 ten 20 ones. Open the last: 0 tens 30 ones. That is 4 ways. Each time one ten becomes 10 ones.

Watch for: Stopping at 3 tens 0 ones and 0 tens 30 ones (2 ways).; Counting 2 tens 10 ones twice.

Investigation (page 5): Swap the digits

Three things can happen. If the tens digit is smaller than the ones digit (14, 26), swapping makes a bigger number (41, 62). If the tens digit is bigger (52, 91), swapping makes a smaller number (25, 19). If the two digits are the same (77), the number stays the same.

Look for: A child who tests several numbers, says what they notice and gives a reason using tens and ones. A number that ends in 0 (30) swaps to 03. A zero on the left does not count, so 03 is just 3 and is smaller: this fits 'tens digit bigger'.

Exit ticket (page 5)

E1. How many sticks are there?

Answer: 48

4 bundles is 40, and 8 singles makes 48.

Watch for: Writing 84.; Writing 12 (4 + 8).

E2. Fill in the blanks for 37.

Answer: 3 tens and 7 ones

The left digit 3 counts the tens, the right digit 7 counts the ones.

Watch for: Swapping: 7 tens and 3 ones.

E3. Write the number: 7 tens and 0 ones.

Answer: 70. The 0 shows there are no ones, and it keeps the 7 in the tens place (without it, it would look like 7 ones).

7 bundles of ten and no singles is 70. The zero holds the ones place.

Watch for: Writing 7.; Writing 700 or 7 0 as 7 + 0.

Sneak-peek puzzles (not in the workbook)

1. 24 stars
2. Yes. 3 bundles of ten is 30, and 4 more is 34.

Answer key: Lesson 2, Number Words and the Number Line

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the missing numbers in the boxes.

Answer: 10 and 12

The line goes 8, 9, 10, 11, 12, 13, 14. The empty boxes are 10 and 12.

Watch for: Writing 9 or 11 again (not counting on).; Leaving a box empty.

S2. Fill in the missing numbers.

Answer: 29; 39 and 41; 70

(a) 28, then 29, then 30. (b) 38, then 39, then 40, then 41: the line goes on after 40. (c) One before 71 is 70.

Watch for: Writing 310 for the number after 38 (the ones digit goes past 9).; Stopping at 40 and leaving the last box empty.

S3. Fill in the number just before and the number just after.

Answer: 34 and 36; 59 and 61; 78 and 80

(a) 34 and 36. (b) Back from 60 is 59, on from 60 is 61. (c) Back from 79 is 78. After 79 the ones are used up, so a new ten begins: 80.

Watch for: Writing 50 and 70 (a whole ten instead of one step).; Writing 710 for the number after 79.

S4. Write the numbers in digits.

Answer: 15, 50, 24

(a) fifteen is ten and 5: 15. (b) fifty is 5 tens: 50. (c) twenty is 20 and 4 more: 24.

Watch for: Writing 50 for fifteen, or 15 for fifty (teen and ty mixed up).; Writing 42 for twenty-four (digits the wrong way round).

S5. Write each number in words. Word bank: twenty, forty, sixty.

Answer: twenty, forty

20 is 2 tens: twenty. 40 is 4 tens: forty (no u).

Watch for: Writing fourty (forty has no u).; Copying the word for a different number, for example sixty for 40.

S6. Fill in the missing numbers. The numbers go up by 10 each time.

Answer: 30, 50 and 80

0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. The missing marks are 30, 50 and 80.

Watch for: Writing 21, 25 or 75 (counting on in ones instead of tens).; Leaving out the mark at 80.

S7. Fill in the blanks. Each number is between two tens.

Answer: 20 and 30; 60 and 70

(a) 23 is 2 tens and 3 more: between 20 and 30. (b) 68 is between 60 and 70.

Watch for: Writing 22 and 24 (the numbers just before and after, not the tens).; Writing 2 and 3 (digits) instead of 20 and 30.

S8. Ben counts: 57, 58, 59, 510, 511.

Answer: 60. After 9 the ones are used up, so a new ten begins. Ben kept counting ones.

After 59 the ones are used up, so we start a new ten: 6 tens and 0 ones, 60. Ben kept adding ones after the 9, so he wrote 510.

Watch for: Writing 510 (the ones digit goes past 9).; Circling Yes (not noticing 510 follows 59).

Think harder (page 3)

T1. Which ten is nearer? Fill in the blanks.

Answer: 50; 90

(a) 47 to 50 is 3 hops, 47 to 40 is 7 hops: 50. (b) 89 to 90 is 1 hop, 89 to 80 is 9 hops: 90.

Watch for: Choosing the ten with the same tens digit (40 for 47) without counting the hops.; Counting the hops from the wrong ten.

T2. How many numbers are there between 28 and 35? (Do not count 28 and 35.)

Answer: 6 (29, 30, 31, 32, 33, 34)

29, 30, 31, 32, 33, 34. That is 6 numbers. The hops from 28 to 35 are 7, but the numbers in the middle are one fewer.

Watch for: Answering 7 (counting the hops, or counting 35).; Answering 8 (counting 28 and 35 as well).

T3. My number is between 50 and 60. It is nearer to 60 than to 50. The number before it is halfway between 50 and 60. What number is it?

Answer: 56

Between 50 and 60 and nearer to 60: 56, 57, 58 or 59. The halfway number is 55. One after 55 is 56, so clue 3 fits only 56.

Watch for: Answering 55 (that is halfway, not nearer to 60).; Answering 54 (one before 55: clue 3 says one before ME is halfway).

T4. Mei's frog is on 90. It hops 2 numbers at a time.

Answer: 96 and 98; No (it lands on 98, then 100)

90, 92, 94, 96, 98, 100. The frog lands on 96 and 98. It hops over 99 (from 98 to 100), so it does not land on 99.

Watch for: Writing 95 and 96 (hopping 1 at a time).; Circling Yes for 99 (counting every number, not every second number).

Go further (page 4)

G1. Complete each line. The jumps are equal.

Answer: (a) 10, 20, jump 10. (b) 15, 20, 25, jump 5. (c) 53, 56, 59, jump 3.

(a) 3 jumps from 0 to 30: 10 each. (b) 4 jumps from 10 to 30, which is 20 altogether: 5 each. (c) 4 jumps from 50 to 62, which is 12 altogether: 3 each.

Watch for: Always using a jump of 10.; Counting the boxes (3) as the jumps (4).

G2. Mei's frog is on 28. Ali's frog is on 52. They hop towards each other, 1 number at a time, at the same time. On which number do they meet?

Answer: 40

From 28 to 52 is 24 hops (28 to 30 is 2, 30 to 50 is 20, 50 to 52 is 2). Each frog makes half: 12 hops. 28 and 12 more is 40. 52 and 12 back is 40. They meet on 40.

Watch for: Choosing 38 or 50, one frog's own number, or adding 28 and 52.; Counting the hops wrongly across the tens.

G3. Make a puzzle. The jumps are equal. Write numbers in 2 boxes that are not next to each other. Leave 3 boxes empty. Give it to a grown-up.

Answer: Any correct line, for example 15, ___, ___, ___, 35 (jump 5: 20, 25, 30).

Example: jump 5: 15, 20, 25, 30, 35. Hide 20, 25 and 30. The grown-up counts the 4 jumps from 15 to 35, which is 20 altogether, so the jump is 5, and the boxes are 20, 25 and 30. Check: the same jump between every two boxes.

Watch for: Making jumps that are not equal.; Giving two numbers that do not fit one equal jump.

Investigation (page 5): Halfway

41, 42, 43 and 44 are nearer to 40. 46, 47, 48 and 49 are nearer to 50. 45 is exactly halfway: 5 hops each way. Rule: ones digit 1 to 4, the lower ten is nearer; ones digit 6 to 9, the higher ten is nearer; ones digit 5, halfway. It works between every two tens, for example 61 to 69.

Look for: A child who tests several numbers by counting hops, notices 45 is special, and says why the ones digit decides it. A child who finds the same pattern between 60 and 70 has generalised.

Exit ticket (page 5)

E1. Write the number in digits.

Answer: 73

Seventy is 70, and 3 more makes 73.

Watch for: Writing 37.; Writing 703.

E2. Fill in the blanks. 76 is between two tens.

Answer: 70 and 80

76 is 7 tens and 6 more, so it is past 70 and not yet 80.

Watch for: Writing 75 and 77 (just before and after, not the tens).; Writing 7 and 8.; Writing 60 and 70.

E3. The frog is on 99. It hops 1 number forward. Where does it land?

Answer: 100

After 99 comes 100, one hundred: ten tens.

Watch for: Writing 910 or 9100.; Writing 90.

Sneak-peek puzzles (not in the workbook)

1. 20 and 23
2. 60. After 59 the ones are used up, so a new ten begins.

Answer key: Lesson 3, Digit Cards

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. You have two digit cards: 3 and 5. Make two numbers with both cards.

Answer: 35 and 53. 53 is greater.

With 3 on the left the number is 3 tens and 5 ones: 35. With 5 on the left it is 5 tens and 3 ones: 53. The same two cards make two numbers. 53 is greater, because 5 tens is more than 3 tens.

Watch for: Writing 8 (3 + 5).; Writing 35 twice.

S2. Write the greater number each time.

Answer: 72, 48, 65, 30

(a) 7 tens is more than 2 tens: 72. (b) Same tens, 8 ones is more than 1 one: 48. (c) 6 tens is more than 5 tens: 65. (d) 3 tens is more than 1 ten: 30.

Watch for: Choosing the number with the greater ones digit (saying 13 is greater than 30).; Not looking at the tens first.

S3. Count the bars and squares in Mei's picture. Ben has 4 bars and 1 square.

Answer: Mei 38, Ben 41, Ben has more

Mei has 3 bars and 8 squares: 38. Ben has 4 bars and 1 square: 41. 4 tens is more than 3 tens, so Ben has the greater number, even though Mei has more squares.

Watch for: Saying Mei has more because she has 8 squares.; Reading Mei's picture as 83.

S4. Make two numbers with both cards 6 and 2.

Answer: 62 and 26

The greater card on the left makes the greater number: 62. The smaller card on the left makes the smaller number: 26.

Watch for: Writing 62 for both.; Writing 8 (6 + 2).

S5. Arrange these in order. Begin with the smallest number: 52, 25, 41, 14.

Answer: 14, 25, 41, 52

14 has 1 ten, 25 has 2 tens, 41 has 4 tens, 52 has 5 tens. Smallest first: 14, 25, 41, 52.

Watch for: Putting 25 after 41 because 5 is greater than 1.; Writing them greatest first.

S6. Kumar says that 19 is greater than 21.

Answer: No. 21 is greater. 2 tens is more than 1 ten.

19 is 1 ten and 9 ones. 21 is 2 tens and 1 one. 2 tens is more than 1 ten, so 21 is greater. Kumar looked at the ones digits first. Look at the tens first.

Watch for: Saying Kumar is right because 9 is more than 2.; Not giving a reason (any clear sentence about the tens is fine).

S7. You have the cards 2 and 8. Use both cards each time.

Answer: 82 and 28

82 has 8 tens, which is more than 5 tens. 28 has only 2 tens, which is less than 5 tens.

Watch for: Writing 82 for both parts.; Writing 8 and 2.

S8. Siti has 25 stickers. Ravi has 52 stickers. Draw bars and squares in the box.

Answer: Ravi (52 is 5 tens and 2 ones; 25 is 2 tens and 5 ones)

Siti: 2 bars and 5 squares. Ravi: 5 bars and 2 squares. 5 bars is more than 2 bars, so Ravi has more stickers, even though both numbers use a 2 and a 5.

Watch for: Saying they have the same because they use the same digits.; Drawing 2 and 5 single squares for Siti.

Think harder (page 3)

T1. You have the cards 3, 5 and 8. Use two different cards to make the greatest number. Then make the smallest number.

Answer: 85 and 35

Numbers you can make: 35, 38, 53, 58, 83, 85. The greatest is 85 (8 tens, the greatest card on the left, then the next greatest card). The smallest is 35.

Watch for: Using all three cards.; Writing 88 (using the 8 twice).

T2. You have the cards 0, 3 and 7. Use two cards to make the smallest and the greatest two-digit numbers.

Answer: 30 and 73

The 0 can not go on the left (03 is just 3). The two-digit numbers are 30, 37, 70 and 73. The smallest is 30 and the greatest is 73.

Watch for: Writing 03 or 3 for the smallest.; Writing 37 for the smallest.

T3. You have the cards 1, 3, 5 and 7. Make two-digit numbers with two different cards. The number is more than 40. Its left digit is greater than its right digit. Write all the numbers that fit.

Answer: 51, 53, 71, 73, 75

More than 40 means a tens card of 5 or 7. With 5 on the left: 51, 53, 57. With 7 on the left: 71, 73, 75. Clue 2 removes 57, because 5 is not greater than 7. Left: 51, 53, 71, 73, 75.

Watch for: Stopping after one or two numbers.; Writing 57 (5 is less than 7).

T4. You have the cards 2, 4 and 6. Use two different cards each time. Write all the two-digit numbers you can make.

Answer: 6 numbers: 24, 26, 42, 46, 62, 64

2 on the left: 24, 26. 4 on the left: 42, 46. 6 on the left: 62, 64. That is 2 + 2 + 2 = 6 numbers.

Watch for: Missing some by listing in a random order.; Writing 22, 44 or 66 (each card can be used only once).

Go further (page 4)

G1. You have the cards 2, 5, 6 and 9. Use two different cards to make a two-digit number.

Answer: 59 and 52

The numbers are 25, 26, 29, 52, 56, 59, 62, 65, 69, 92, 95, 96. Less than 60: 25, 26, 29, 52, 56, 59, so the greatest is 59. Greater than 50: 52, 56, 59, 62, 65, 69, 92, 95, 96, so the smallest is 52.

Watch for: Writing 96 or 69 for part a (they are greater than 60).; Writing 25 for part b (it is not greater than 50).

G2. You have the cards 1, 2, 3 and 4. Make two-digit numbers. How many numbers can you make?

Answer: 12 and 16

Each card once: 4 choices for the tens card, then 3 choices for the ones card, so 4 x 3 = 12. A card may be used twice: 4 choices and 4 choices, so 16 (the 12 and the four numbers 11, 22, 33, 44).

Watch for: Counting only 8 or 6 numbers.; Forgetting 11, 22, 33 and 44 in part b.

G3. I make a two-digit number with two different cards from 1, 3, 4 and 7. My number is between 35 and 45. My ones digit is greater than my tens digit. What number am I?

Answer: 37

The numbers are 13, 14, 17, 31, 34, 37, 41, 43, 47, 71, 73, 74. Between 35 and 45: 37, 41, 43. Ones digit greater than tens digit: 37 (7 is greater than 3). 41 and 43 have a ones digit that is smaller.

Watch for: Stopping at 41 or 43 and not checking the ones digit.; Writing 34 or 47 (they are not between 35 and 45).

Investigation (page 5): Is the greater card always first?

For two different cards, the greater card on the left always makes the greater number, because the tens decide. With the cards 0 and 8 you can only make 80, because 08 is just 8. With the same card twice (4 and 4) you can only make 44. So you can make two different two-digit numbers only when the two cards are different and neither is 0.

Look for: A child who tests several pairs, notices that the tens digit decides, and finds the two special cases (a 0 card, two matching cards). A reason using tens and ones is better than "because it is greater".

Exit ticket (page 5)

E1. Write the greater number: 39 or 41.

Answer: 41

41 has 4 tens and 39 has only 3 tens, so 41 is greater. Look at the tens first.

Watch for: Writing 39 because 9 is greater than 1.

E2. Make two numbers with both cards 7 and 3.

Answer: 73 and 37

The greater card on the left makes the greater number: 73. The smaller card on the left makes the smaller number: 37.

Watch for: Writing 73 for both.; Writing 10 (7 + 3).

E3. Arrange these in order. Begin with the smallest number: 63, 36, 60.

Answer: 36, 60, 63

36 has 3 tens. 60 and 63 both have 6 tens. 60 has 0 ones and 63 has 3 ones, so 60 comes before 63.

Watch for: Putting 60 before 36 or after 63.; Writing them greatest first.

Sneak-peek puzzles (not in the workbook)

1. 18
2. Mei. Her number is 84, which is 8 tens. Ali has 48, which is only 4 tens.

Answer key: Lesson 4, The 100 Square Explorer

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Colour 10, 20, 30 and so on to 100 as you count in 10s. Fill in the blanks.

Answer: They end in 0. They make a line going down.

The numbers are 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Every one ends in 0. Each is 10 more than the one before, which is one row down, so they make a straight line down the last column.

Watch for: Colouring every number in the last row (91 to 100).; Saying the line goes across: the colours are one under the other, so it goes down.

S2. Count in 5s. Colour 5, 10, 15 and keep going to 100. Fill in the blanks.

Answer: They end in 5 or 0. They make 2 lines going down.

The numbers are 5, 10, 15, 20, and on to 100. They end in 5 or 0. The 5s sit in the fifth column and the 10s in the last column, so there are two lines going down.

Watch for: Skipping a number, such as 35 or 45.; Saying 1 line because the tens are in a line too (there are 2 lines: the 5s and the 0s).

S3. What are the next two numbers in each of these patterns?

Answer: 10, 12 | 30, 35 | 60, 70 | 41, 51

(a) Counting in 2s: 8, then 10, 12. (b) Counting in 5s: 25, then 30, 35. (c) Counting in 10s: 50, then 60, 70. (d) Each step is 10 more, one row down: 31, then 41, 51.

Watch for: (d) writing 32 and 33: the jump is 10, not 1.; Changing the jump half way along.

S4. Count in 2s. Colour 2, 4, 6 and so on to 30. Look at the last digits.

Answer: 2, 4, 6, 8, 0 (any order)

The coloured numbers are 2, 4, 6, 8, 10, 12 and on to 30. Their last digits are 2, 4, 6, 8, 0, and then the same five again. The colours make stripes.

Watch for: Colouring 1, 3, 5 as well.; Writing 5 or 1 as a last digit.

S5. Fill in the missing number in each empty square.

Answer: a 34, b 43, c 55

(a) 34 is between 33 and 35. (b) 43 is under 33, which is 10 more than 33, and above 53. (c) 55 is under 45 (10 more) and between 54 and 56.

Watch for: Writing 44 for b: 44 is next to 43, not under 33.; Adding 1 when moving down instead of 10.

S6. Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it.

Answer: 26 is wrong. She should have said 25.

Counting in fives, 20 is followed by 25, then 30. 26 ends in 6, but numbers counted in fives end in 5 or 0. So Siti should have said 25.

Watch for: Circling 30 or 35.; Writing 26 again instead of 25.

Think harder (page 3)

T1. Put a counter on 27 on your 100 square. Move the counter down 3 rows. Which number does it land on? Write how you know on the lines.

Answer: 57 (each row down is 10 more, so 3 rows is 30 more)

Each row down adds 10, so 3 rows add 30. 27 + 30 = 57. Check on the square: 37, 47, 57.

Watch for: Writing 30 or 33 (adding 3 instead of 3 tens).; Counting 3 squares along the row.

T2. A counter is on 14. It moves down 2 rows. Then it moves right 3 squares. Which number does it land on? Write each step on the lines.

Answer: 37 (14, 24, 34, then 35, 36, 37)

Down 2 rows is 20 more: 14, 24, 34. Right 3 squares is 3 more: 35, 36, 37. It stops on 37.

Watch for: Adding 2 and 3 to 14 (19).; Going down by 2 squares instead of 2 rows (16).

T3. Look at this pattern: 6, 7, 9, 12, 16 and so on. Find the next two numbers.

Answer: The gaps are 1, 2, 3, 4. The next numbers are 21 and 27.

The gaps are 1, 2, 3, 4: each gap is 1 bigger. The next gap is 5, so 16 + 5 = 21. The gap after is 6, so 21 + 6 = 27.

Watch for: Adding 4 each time (20, 24).; Finding the gaps but forgetting to use them to go on.

T4. Mei counts in 2s up to 30: 2, 4, 6, 8 and so on. Ali counts in 5s up to 30: 5, 10, 15, 20 and so on. Write the numbers they both say.

Answer: 10, 20 and 30

Mei says 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30. Ali says 5, 10, 15, 20, 25, 30. Both say 10, 20 and 30.

Watch for: Writing only 10.; Including 5, 15 or 25: Mei does not say numbers ending in 5.

Go further (page 4)

G1. Here is a number chain: 2, 6, _ _ _ _ _ _ _, 14, 22, _ _ _ _ _ _ _, 58. One rule makes the whole chain.

Answer: 8 and 36 (2 + 6 = 8, 14 + 22 = 36, and 22 + 36 = 58)

Each number after the first two is the two numbers before it added together. 2 + 6 = 8, so the first blank is 8. Check: 6 + 8 = 14 and 8 + 14 = 22. The last blank is 14 + 22 = 36, and 22 + 36 = 58 matches the end of the chain.

Watch for: Adding 6 each time (8, 14, 20).; Filling the first blank but not the last.; Not checking the rule on the numbers at the end of the chain.

G2. Look at this list: 5, 50, 10, 45, 15, 40 and so on.

Answer: The next two numbers are 20 and 35. The 10th number is 30.

Numbers 1, 3, 5 go up by 5: 5, 10, 15, then 20. Numbers 2, 4, 6 go down by 5: 50, 45, 40, then 35. The list goes on 20, 35, 25, 30. The 10th number is 30.

Watch for: Treating the list as one pattern (adding or taking away the same amount).; Getting the 8th number right and the 10th wrong.

G3. Look at this block of 4 squares.

Answer: Both add up to 79, and 52 + 63 and 53 + 62 both make 115. It always works: 1 and 10 are added in both pairs.

If the top left number is a, the others are a + 1, a + 10 and a + 11. One slant is a + (a + 11) and the other is (a + 1) + (a + 10). Both are the same: 2 a and 11. For the first block both are 79.

Watch for: Only trying one block and saying it always works.; Adding the numbers across the rows instead of along the slants.

Investigation (page 5): Counting in 3s

Counting in 3s makes slanting lines going down and to the left, not a straight column, because a row has 10 squares and 10 is not said when we count in 3s. The picture repeats every 3 rows: 33, 36 and 39 sit under 3, 6 and 9.

Look for: A child who colours carefully, says what shape they see and gives a reason that uses rows of ten. Any clear noticing counts (zig-zag, slanting, it moves left each row). Colouring to 60 shows the repeat. Do not name the word multiple.

Exit ticket (page 5)

E1. Count in 5s. What are the next two numbers?

Answer: 50, 55

Counting in fives: 45, then 50, then 55.

Watch for: Writing 46 and 47.; Counting in tens (55 and 65).

E2. A counter is on 26 on your 100 square. First it moves down 1 row. Then it moves right 1 square.

Answer: 36, then 37

Down one row is 10 more: 36. Right one square is 1 more: 37.

Watch for: Writing 27 for down.; Writing 46 (adding 20).

E3. Mei counts in 2s: 2, 4, 6, 8 and so on. Will she say 25?

Answer: No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8.

Counting in twos, the last digit is always 0, 2, 4, 6 or 8. 25 ends in 5, so Mei does not say 25.

Watch for: Saying yes because 25 is next to 24 or 26.

Sneak-peek puzzles (not in the workbook)

1. 25 and 30
2. 46. Down one row is 10 more, because a row has ten squares.

Answer key: Lesson 5, Make-a-Ten Adding

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the ten-frames. Fill in the blanks.

Answer: 1 more; 10 and 3; 9 + 4 = 13

The first frame has 1 empty space, so 9 needs 1 more. Move 1 counter from the 4. The first frame is full (10) and 3 are left. 10 + 3 = 13.

Watch for: Writing 10 and 4: forgetting to take the moved counter away from the 4.; Counting all the counters again and slipping (12 or 14).

S2. Draw 7 dots and 6 dots in the box. Ring 10 dots. Then fill in the blanks.

Answer: 7 + 6 = 10 + 3 = 13 (ten dots ringed, 3 left over)

Ring the 7 dots and 3 of the 6 dots to make ten. The other 3 dots are outside the ring. So 7 + 6 = 10 + 3 = 13.

Watch for: Ringing ten dots but forgetting the dots left outside.; Miscounting when counting every dot one by one.

S3. Fill in the missing numbers to make the next ten.

Answer: 3, 2, 4, 1

(a) 7 and 3 make 10. (b) 18 is 1 ten and 8 ones, and 8 and 2 make 10, so 18 + 2 = 20. (c) 6 and 4 make 10, so 26 + 4 = 30. (d) 9 and 1 make 10, so 39 + 1 = 40.

Watch for: Writing the next ten itself (20 for part b).; Counting from the tens digit instead of the ones digit.

S4. Add 28 + 6. Fill in the blanks.

Answer: 2 more; 6 = 2 and 4; 28 + 6 = 34

28 needs 2 more to make 30. Split 6 into 2 and 4. 28 + 2 = 30, and 30 + 4 = 34.

Watch for: Writing 36: adding all 6 after reaching 30, when 2 of the 6 were already used.; Splitting 6 into 2 and 3 or another wrong pair.

S5. Add 35 + 7. Fill in the blanks.

Answer: 5 more; 35 + 7 = 42

35 needs 5 more to make 40. 7 is 5 and 2. 35 + 5 = 40, and 40 + 2 = 42.

Watch for: Writing 47 (adding 7 to the 4 tens).; Forgetting the 2 left over after the ten.

S6. Add.

Answer: 15, 23, 32, 3 more, 53

(a) 9 + 1 = 10, 10 + 5 = 15. (b) 18 + 2 = 20, 20 + 3 = 23. (c) 27 + 3 = 30, 30 + 2 = 32. (d) 47 needs 3 more to make 50, 6 is 3 and 3, so 47 + 3 = 50 and 50 + 3 = 53.

Watch for: Moving the right amount to make the ten but then adding the whole second number.; Making a ten with the wrong number (for example 9 + 6 = 9 + 4 + 2).

S7. Siti adds 8 + 6. Here is her working: 8 + 2 = 10, then 10 + 6 = 16. Answer (a), (b) and (c).

Answer: Wrong. The right answer is 14. 10 + 6 should be 10 + 4: she moved 2 from the 6, so only 4 are left.

8 + 2 = 10 uses 2 of the 6. Only 4 are left, so the sum is 10 + 4 = 14. Siti added all 6 again.

Watch for: Saying she is right because 10 + 6 looks like a good sum.; Writing 14 without saying what she forgot (a spoken reason is fine).

Think harder (page 3)

T1. Mei adds 4 + 9 + 6. Fill in the blanks.

Answer: 4 and 6 make 10. The total is 19.

4 and 6 make 10. 10 + 9 = 19. Check the long way: 4 + 9 = 13, and 13 + 6 = 19.

Watch for: Adding the first two numbers (4 + 9) and stopping or slipping.; Thinking the pair must sit side by side.

T2. Add 8 + 5 + 2 + 7. Show your working on the lines.

Answer: 22 (8 + 2 = 10, 5 + 7 = 12, 10 + 12 = 22)

8 + 2 = 10. 5 + 7 = 12 (5 + 5 + 2). 10 + 12 = 22.

Watch for: Adding from left to right and losing track.; Using a number twice or leaving one out.

T3. Circle the sum that does not make 30. Then write how you know.

Answer: 19 + 12 (it makes 31). The ones 9 and 2 make 11, not 10.

7 + 3 = 10, 8 + 2 = 10 and 6 + 4 = 10, so those make 30. 9 + 2 = 11, so 19 + 12 is 31.

Watch for: Adding every sum one by one and slipping.; Choosing 18 + 12 because 18 looks big.

T4. Ben has 18 red marbles, 34 blue marbles and 12 green marbles. How many marbles does Ben have altogether? Show your working on the lines.

Answer: 64 (18 + 12 = 30, 30 + 34 = 64)

18 + 12 = 30. 30 + 34 = 64. The order does not change the total.

Watch for: Leaving out one group of marbles.; Adding 18 + 34 first and slipping on the ten.

Go further (page 4)

G1. Add 18 + 7 + 12 + 23 + 3 + 17. Then change one number. Show your working on the lines.

Answer: (a) 80 (18 + 12 = 30, 7 + 3 = 10, 23 + 17 = 40). (b) 13: 80 + 10 = 90, so the 3 changes to 13.

(a) 18 + 12 = 30. 7 + 3 = 10. 23 + 17 = 40. 30 + 10 + 40 = 80. Another way: 7 + 23 = 30, 18 + 12 = 30, 3 + 17 = 20, total 80.
(b) The total must go up from 80 to 90, which is 10 more. The 3 must become 3 + 10 = 13.

Watch for: Adding left to right and losing the running total.; Using a number twice or leaving one out.; Changing the 3 to 10 (the total goes up by 7, not 10).

G2. Look at this sum: 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1. Fill in a and b.

Answer: 45 and 55 (four pairs of 10, and the 5 is left)

1 + 9, 2 + 8, 3 + 7 and 4 + 6 make four tens: 40. The 5 has no partner. 40 + 5 = 45. Adding 10 makes 55.

Watch for: Forgetting the 5 in the middle has no partner.; Making only three pairs.

G3. Write a sum of 6 numbers that makes 100. Many answers are right. Show your working on the lines.

Answer: Any six numbers that total 100. For example 45 + 5 + 20 + 17 + 3 + 10 = 100 (45 + 5 = 50, 17 + 3 = 20, 20 + 10 = 30).

Choose pairs first: 50 + 20 + 30 = 100. Then split each into two numbers, for example 45 + 5, 17 + 3, 20 + 10. Add them to check.

Watch for: Choosing numbers that total 90 or 110: add every number to check.; Using fewer than 6 numbers.

Investigation (page 5): Move the fewest

Fill up the bigger number: it has the fewest empty spaces, so you move the fewest dots. The number moved is 10 minus the bigger number. 9 + 6: move 1, not 4. 8 + 7: move 2, not 3. If the two numbers are the same (7 + 7), either way moves 3.

Look for: A child who tries several pairs, records how many dots moved each way, and notices the bigger number is closer to ten. Credit any clear reason, such as the bigger number has fewer empty spaces.

Exit ticket (page 5)

E1. Add.

Answer: 15

8 + 2 = 10. 7 is 2 and 5, so 10 + 5 = 15.

Watch for: Writing 18 (10 + 8).; Writing 14 or 16 after counting on fingers.

E2. Add.

Answer: 43

38 + 2 = 40. 5 is 2 and 3, so 40 + 3 = 43.

Watch for: Writing 45 (40 + 5).; Writing 33 or 83.

E3. Add 6 + 9 + 4. Fill in the blanks.

Answer: 19. 6 and 4 make 10, then 10 + 9 = 19.

6 + 4 = 10. 10 + 9 = 19.

Watch for: Writing 6 and 9 as the pair that makes 10 (6 + 9 is 15).; Not noticing that 6 and 4 make a ten.

Sneak-peek puzzles (not in the workbook)

1. 12
2. Yes. Nothing was added or taken away. 9 + 5 = 14 and 10 + 4 = 14.